A Project Report

On

Genome Analysis Using Graphs and Machine Learning

BY

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(2023A7PS0055H)

Under the supervision of

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SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS OF

CS F367: LABORATORY PROJECT



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI (RAJASTHAN)
HYDERABAD CAMPUS

(OCTOBER 2026)

Acknowledgements

I would like to thank Prof. Apurba Das for giving me the chance to work on this project and for guiding me throughout. His suggestion to try other graph embedding methods and to look for more features shaped most of the work in this report. I also thank BITS Pilani, Hyderabad Campus for giving access to the high performance computing cluster, on which all the experiments were run. Finally, I thank the authors of SCI and of the GM12878 Hi-C study for making their code, container and data publicly available, without which this work would not have been possible.



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Certificate

This is to certify that the project report entitled "**Genome Analysis Using Graphs and Machine Learning**" submitted by Mr. Harshal Shah (ID No. 2023A7PS0055H) in partial fulfillment of the requirements of the course CS F367, Laboratory Project Course, embodies the work done by him/her under my supervision and guidance.

Date: **(Prof. Apurba Das)**
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Abstract

In this project we study how the genome is folded inside the nucleus, using Hi-C contact data, graphs and machine learning. Our starting point is SCI (Ashoor et al., 2020), a method that builds a graph from contacts between different chromosomes, learns a vector for every genome region using the graph embedding method LINE, and then groups these vectors into five sub-compartments using K-means. The aim was to reproduce SCI and then check whether other graph embedding methods give better results.

We first ran SCI on the demo dataset provided by the authors. The graph we got had the same size as the one built by their code (26,977 nodes and 8.35 million edges), and SCI's neural network classifier gave accuracies close to the ones reported in the paper (0.721 vs 0.73 and 0.681 vs 0.68). After this we built the full pipeline on the GM12878 Hi-C data from Rao et al. (2014), and compared LINE with weighted DeepWalk, HOPE and a combination of DeepWalk and HOPE. All results were scored against Rao's sub-compartment labels and against SCI's published labels, and were also checked using chromosome identity, resampling of chromosomes and replication timing data.

The main findings are as follows. SCI's way of using LINE, where the two orders are trained separately and then joined together, gives clusters that are close to random on the full data. This happens because the order-2 vectors are about 70 times longer than the order-1 vectors, and the authors' own code shows the same problem. HOPE-based methods, together with a simple correction to the contact map that we call pair-O/E, give the best agreement with Rao's labels, reaching an ARI of 0.633 compared to 0.503 for the published SCI labels. However, the improvement is not very large. Over an improved version of LINE the gain is only about 0.06 to 0.11 ARI, and on replication timing our best results only match Rao's labels and do not beat them. We were also not able to reproduce SCI's published labels, and even the authors' own container does not give back its own saved demo result. In the next part of the project we will design a data structure and a query processor for this type of genome data.

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1 Introduction

1.1 Background

DNA inside the nucleus is not stretched out in a straight line. It is folded in a complex way, and because of this folding, regions that are very far apart along the DNA sequence can end up physically close to each other. This matters because the 3D arrangement of the genome affects which genes are active and which are not. Studying which parts of the genome come in contact with each other therefore helps in understanding how genes are regulated.

1.2 Hi-C and genomic sub-compartments

Hi-C is an experimental technique that measures these contacts across the whole genome in one go. The genome is divided into equal sized pieces called bins, and the output is a contact matrix in which each entry tells how many times two bins were found in contact. At a broad level the genome can be divided into two compartments, A and B. The A compartment is mostly open and active, while the B compartment is mostly closed and inactive. Rao et al. (2014) produced a high resolution Hi-C map of the GM12878 cell line and further divided these compartments into six sub-compartments, namely A1, A2, B1, B2, B3 and a very small B4. Each of these has its own pattern of histone marks and replication timing. Their labels are the main reference used in this work.

1.3 SCI

SCI (Ashoor et al., 2020) is a different approach for finding sub-compartments. It only uses contacts between different chromosomes. These contacts are converted into a graph where every bin is a node, the graph embedding method LINE is used to learn a vector for each node, and K-means is then used to cluster these vectors into five groups (C1 to C5). According to the paper, SCI's labels are better than Rao's on some measures such as how compact the clusters are.

1.4 Motivation

Graph embedding is a large area and LINE is only one of the many methods available. The SCI paper did compare LINE with DeepWalk and HOPE, but only using scores that do not depend on any reference labels. Our supervisor asked us to reproduce SCI, try out other graph embedding algorithms and see whether they perform better. While working on this we came across some issues in the original method, so a significant part of the work ended up being about understanding why SCI behaves the way it does and whether its published labels can actually be reproduced.

1.5 Research questions

The work in this report tries to answer the following questions:

1. Can SCI be run and its results reproduced, first on the demo dataset and then on the full GM12878 data?
2. How does SCI's LINE setup behave on the real data, and what is the reason for it?
3. Do other graph embedding methods (weighted DeepWalk, HOPE and their combination) agree better with the known sub-compartments than LINE does?

4. Does a simple correction for differences in contact levels between chromosome pairs (pair-O/E) improve the results?
5. Do the new clusterings also separate independent biological signals like replication timing, or do they just copy Rao's labels?
6. Is it possible to regenerate the published SCI labels?

2 Related Work

2.1 Rao et al. (2014)

Rao et al. used in situ Hi-C to map the GM12878 genome at kilobase resolution. They identified sub-compartments by clustering the interchromosomal contact matrix using a Gaussian hidden Markov model. They also showed that every sub-compartment has its own set of chromatin marks and replication timing. A1 and A2 are gene rich and replicate early, B1 is associated with the Polycomb mark H3K27me3, and B2 and B3 replicate late and lie close to the nuclear periphery and the nucleolus. B4 is a small group that is found only on chromosome 19. These labels are publicly available and we use them as our reference.

2.2 SCI

SCI (Ashoor et al., 2020) constructs a graph from interchromosomal contacts, embeds it using LINE and clusters it with K-means ($k = 5$). In the paper, LINE was compared with DeepWalk and HOPE on silhouette and centrality scores, and LINE was chosen. The authors supported their labels using silhouette scores, ChIA-PET loop ratios and a deep neural network (sci-DNN) that predicts the labels from other genomic features. The code, a Docker container, a demo dataset and the published GM12878 labels are all publicly available. The paper however does not report how well its labels agree with Rao's labels in terms of ARI or NMI.

2.3 LINE

LINE (Tang et al., 2015) learns node vectors directly from the edges of a graph. First-order proximity (order 1) tries to keep nodes that are directly connected close to each other, while second-order proximity (order 2) tries to keep nodes with similar neighbours close. Edges are sampled with probability proportional to their weight, and negative sampling is used during training. In the original LINE toolkit, both orders are trained separately, each one is normalised, and only then are they joined.

2.4 DeepWalk

DeepWalk (Perozzi et al., 2014) performs short random walks on the graph and treats each walk like a sentence. Node vectors are then learned using word2vec (Mikolov et al., 2013), so nodes that frequently appear together in walks end up with similar vectors. In the weighted version, the next node in a walk is chosen with probability proportional to the edge weight.

2.5 HOPE

HOPE (Ou et al., 2016) is based on matrix factorisation. It builds a proximity matrix for all pairs of nodes and reduces it using singular value decomposition (SVD). We use the Katz proximity (Katz, 1953), which

counts paths of all lengths between two nodes while giving less weight to longer paths.

These three methods were chosen because each one is based on a different idea: edges (LINE), random walks (DeepWalk) and matrix factorisation (HOPE). Methods like SNIPER (Xiong and Ma, 2019) are trained on Rao's labels, which makes them supervised, so they cannot be directly compared with our unsupervised results.

3 Experimental Design

3.1 Overall pipeline

The pipeline used in this project is shown in Figure 1. All the steps after building the graph are kept the same for every method, so any difference in results comes only from the graph and the embedding method used.

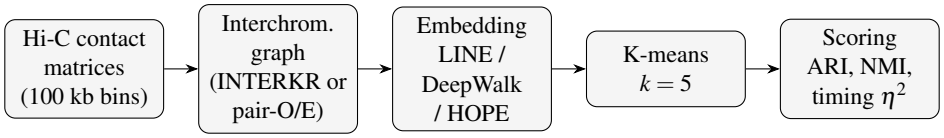


Figure 1: The pipeline used for every method.

For clustering we used K-means with $k = 5$, with 100 restarts and three different random seeds. Out of these, the solution with the lowest K-means inertia (the value that K-means tries to minimise) was kept. This way of choosing does not use any labels. With 100 restarts, the results from all three seeds agreed with each other to within about ± 0.001 ARI for every method except concatenated LINE, whose embedding is close to noise (Section 6.8).

3.2 Evaluation strategy

Since no single score gives the full picture, we used a few different kinds of scores:

- **Agreement with reference labels.** ARI (adjusted Rand index; Hubert and Arabie, 1985) and NMI (normalised mutual information) measure how similar two labellings are. Both are close to 0 for random labels and equal to 1 when the labels are identical. Every clustering was compared with Rao's labels as well as the published SCI labels.
- **Internal scores.** Silhouette (Rousseeuw, 1987; higher is better) and Davies–Bouldin (Davies and Bouldin, 1979; lower is better) tell how compact and well separated the clusters are, without using any labels.
- **Chromosome check.** NMI between the clusters and the chromosome each bin belongs to. A high value means that the clusters are partly just following chromosomes. Rao's own labels give a value of 0.100 here, since some chromosomes genuinely have more of certain sub-compartments, so the target is not zero.
- **Biological signals.** Replication timing and histone marks from ENCODE (ENCODE Project Consortium, 2012), scored using η^2 (Section 8).

All the scores are computed on the same set of 26,226 bins (Section 4.2).

3.3 Statistical testing

Bins that lie on the same chromosome are not independent of each other, so resampling individual bins would give confidence intervals that are too narrow. Instead, we resampled whole chromosomes. We picked 22 chromosomes with replacement, recalculated the score using the bins from those chromosomes, and repeated this 1,000 times to get 95% percentile intervals. For comparing two methods, paired differences were taken on the same resamples. A difference was considered clear only when its interval did not contain zero.

We also kept a few rules to make sure we were not fooling ourselves. No setting was tuned against Rao's labels. Every new idea and the test for it was written down before running it, and it was run only once. All decisions were recorded in a log file as the work progressed.

4 Dataset and Graph Construction

4.1 GM12878

The data used is the combined GM12878 interchromosomal contact matrices from Rao et al. (2014), available on GEO under accession GSE63525. GM12878 is a human lymphoblastoid cell line, and it is the same cell line used by Rao and by the SCI paper. The data is quite large, and a good part of the practical work went into handling it. The main files are listed in Table 1. The archive contains all resolutions, two read quality filters (MAPQ$\geq$30 and MAPQ$\geq$0) and multiple normalisation vectors for each chromosome pair. Instead of unpacking the whole archive, we extracted only the part that was needed (100 kb, MAPQ$\geq$30, autosomes).

Table 1: Main files and their sizes.

File	Size
Interchromosomal archive, compressed (all resolutions)	36 GB
Same archive, uncompressed	224 GB
Extracted part used by us (100 kb, MAPQ$\geq$30, autosomes; 231 pairs, 3,465 files)	5.82 GB
Our graph, INTERKR normalisation (edge list)	8.6 GB
Graph from unfiltered reads (MAPQ$\geq$0)	11 GB
Same graph in SCI's 7-column input format (gzipped)	2.7 GB
SCI authors' demo input file (8,913,782 lines)	490 MB
Graph built from the demo by SCI's code	338 MB
One 100-dimensional embedding (26,743 nodes, text)	25 MB
SCI authors' container image	3.0 GB
Replication timing track (Repli-seq)	10 MB
Histone tracks: H3K27me3 / H3K9me3 / H3K4me3	520 / 663 / 434 MB

4.2 Rao reference labels

Rao's labels cover 28,823 bins of 100 kb each: A1 4,006, A2 5,814, B1 3,494, B2 4,357, B3 8,555, B4 110, and 2,487 bins without any label. The published SCI labels (C1–C5) cover 26,226 bins, which is exactly the same set as Rao's A1–B3 bins. For this reason all scoring is done on these 26,226 bins. On this set, the published SCI labels agree with Rao's labels with **ARI 0.503 and NMI 0.567**, and this is the number we compare against in the rest of the report.

4.3 100-kb binning

The 22 autosomes were divided into bins of 100,000 bases each, giving a total of 28,823 bins. We followed SCI's code exactly here. The bin index is position//100000 + 1, and the order of the nodes is taken from SCI's own chromosome size file (in which chromosome 20 comes before 19 and chromosome 22 comes before 21). This ensures that our node IDs are the same as SCI's.

4.4 Interchromosomal graph

Every bin is a node in the graph. If two bins lying on different chromosomes have a contact value of at least 1, an edge is added between them with weight $\ln(\text{value})$, and the edge is written in both directions. Like SCI, we only use contacts between chromosomes, since contacts within a chromosome mostly depend on the distance along the DNA and hide the pattern we are interested in. We wrote our own graph builder following these rules and checked it against SCI's code. One small difference should be noted here. The paper describes the edge weight as the normalised Hi-C read count and does not mention a threshold, while SCI's code keeps only contacts with value ≥ 1 and uses $\ln(\text{value})$ as the weight. We followed the code, since that is what produces SCI's results.

4.5 INTERKR

The SCI paper does not mention which normalisation was used. We chose Rao's KR normalisation for interchromosomal data (INTERKR; Knight and Ruiz, 2013), in which each raw count is divided by the product of the normalisation values of the two bins. The resulting graph turned out to be very dense, with about 58% of all possible interchromosomal bin pairs having an edge. Out of the 255 million contact entries, 227.5 million satisfy the condition value ≥ 1.

4.6 Pair-O/E

The average level of contact is not the same for all chromosome pairs. Small chromosomes that are rich in genes (16, 17 and 19–22) are in contact with each other more often than other pairs (Lieberman-Aiden et al., 2009). We suspected that this could make the graph reflect which chromosome a bin is on, rather than which sub-compartment it belongs to. So we built a second graph in which every chromosome-pair block is divided by its own mean over valid bins (observed/expected), after which the same edge rule is applied. The reasoning and results for this are given in Section 7, and Table 2 compares the two graphs.

Table 2: The two main graphs. Both have 26,743 nodes with at least one edge.

Graph	Undirected edges	Directed lines written
INTERKR	227,549,310	455,098,620
INTERKR + pair-O/E	142,229,814	284,459,628

5 Reproduction of SCI

5.1 Demo dataset: initial results

As a first step, we ran SCI on the demo dataset that comes with the authors' code, on our own system using their Docker setup. The demo input is 490 MB in size (8,913,782 contact lines). SCI's code converted it into a 338 MB graph with 16,704,136 lines, which corresponds to 8,352,068 edges on 26,977

nodes. This is the same graph that the authors' code builds from this file, so our setup matched theirs at this stage.

Using this demo graph, we did a first comparison of LINE and DeepWalk:

- **LINE:** order-1 and order-2 embeddings of 100 dimensions each, concatenated into 200 dimensions and then clustered with K-means.
- **Unweighted DeepWalk:** random walks where every neighbour has the same chance of being picked, followed by word2vec (skip-gram) with 100 dimensions and then K-means.
- **Weighted DeepWalk:** random walks where the next node is picked with probability proportional to the Hi-C edge weight, so stronger contacts are followed more often, followed by the same word2vec and K-means steps.

All three embeddings had the same 26,977 nodes. The internal scores are shown in Table 3. LINE gave five clusters of nearly equal size (5,341, 5,290, 5,412, 5,418 and 5,516 bins), whereas weighted DeepWalk gave clusters of more varied sizes (3,388, 5,969, 7,407, 5,338 and 4,875 bins).

Table 3: Initial results on the authors' demo graph (26,977 nodes). Internal scores only.

Method	Silhouette ↑	Davies–Bouldin ↓
LINE (orders 1+2 concatenated)	0.0059	9.09
Unweighted DeepWalk	0.007	8.36
Weighted DeepWalk	**0.031**	**3.73**

At this point weighted DeepWalk seemed clearly better than LINE, and using the Hi-C weights in the random walks looked like a promising direction. The plan was to check this against Rao's labels using ARI and NMI and to include HOPE as well. Later on, two things changed the way we looked at these numbers. First, the demo file turned out to be a random 3% sample of the full data (Section 8.5), so it is not suitable for drawing biological conclusions. Second, on the full data, silhouette turned out to be a poor indicator of agreement with Rao's labels (Section 6.8). So these results confirmed that the pipeline was working, but the comparison between methods had to be done again on the full data.

5.2 Real GM12878

Next, the pipeline was run on the full GM12878 data using the INTERKR graph (Table 2). This graph is much bigger than the demo graph, with 227.5 million edges compared to 8.35 million. It has 26,743 nodes with at least one edge, which is 517 more than the 26,226 bins that have published labels. LINE was run with the same settings as in the paper, that is 100 dimensions per order, 5 negative samples and 25 million samples.

5.3 sci-DNN

The SCI repository also has sci-DNN, a neural network that predicts the sub-compartment label of a bin from other genomic features. The paper uses it to show that SCI labels are easier to predict than Rao's HMM labels. When we ran it, we got an accuracy of 0.721 for predicting SCI labels and 0.681 for predicting Rao's labels. The values reported in the paper are 0.73 and 0.68, so this part was reproduced well.

5.4 Reproduction results

The demo pipeline and sci-DNN were reproduced successfully, but the main LINE result on the full data was not. Using the paper's settings, our clusters agree with Rao's labels with an ARI of only 0.049, and with the published SCI labels with an ARI of 0.041, which is close to random. The next section explains the reason for this, and Section 8.4 describes our attempts to regenerate the published labels.

6 LINE and Alternative Graph Embeddings

6.1 SCI's LINE recipe

In SCI, LINE order 1 and order 2 are trained separately, the two vectors are concatenated, the combined vector is normalised and then clustered. We refer to this as the concatenated mode. We used SCI's own modified version of LINE, with the same settings as the authors' defaults (25 million samples, 100 dimensions per order and 5 negative samples).

6.2 Order 1 against order 2

To understand why the concatenated mode fails, we clustered each order separately:

- order 1 on its own gives an ARI of 0.377 against Rao, so it does carry useful signal;
- order 2 on its own gives an ARI of 0.006, which is basically noise;
- the order-2 vectors are around 70 times longer than the order-1 vectors (median length 0.70 compared to 0.010).

Because of this, when the two parts are joined without normalising each one first, the distances between bins are almost entirely decided by the order-2 part, and the useful order-1 part gets ignored.

6.3 Per-order normalisation

The fix for this is to normalise each order to unit length before joining them, which is also how the two orders are combined in the original LINE toolkit. With this change the ARI becomes 0.382 at 25 million samples and 0.443 at 1000 million samples. At 25 million samples it behaves almost the same as order 1 alone (the silhouette and Davies–Bouldin values match up to four decimal places), as the order-2 part hardly adds anything. With more samples order 2 does start to carry some information, but it also starts following chromosomes more closely, as its NMI with chromosome went up from 0.283 at 250 million samples to 0.451 at 1000 million.

6.4 Joint LINE

SCI's LINE code also has a joint mode (order 3), in which a single vector is trained with both objectives together. This avoids the complete collapse (ARI 0.409 on INTERKR), but two of its five clusters contain only 47 bins in total, which makes the score look better than it really is. If only the 26,226 labelled bins are clustered, the ARI drops to 0.299. When we tried the joint mode with 1000 million samples, it crashed every time (four runs across both graphs), so it is not included in the results.

6.5 Weighted DeepWalk

Since the graph is too big for common graph libraries, we wrote our own version of weighted DeepWalk that samples the walks directly from the sparse matrix. The settings used were a walk length of 40, 5 walks per node, 100 dimensions, a window size of 5, skip-gram, 5 epochs and a fixed seed.

6.6 HOPE

HOPE was implemented following the GEM library (Goyal and Ferrara, 2018), using the Katz proximity

$$S = (I - \beta A)^{-1} \beta A,$$

where A is the weighted adjacency matrix. A rank-50 SVD was used, which gives 100 dimensions after joining the source and target vectors. The Katz series converges only if β is less than $1/\rho(A)$, where $\rho(A)$ is the spectral radius of A. The commonly used default of $\beta = 0.01$ is too big for this dense graph and the series does not converge, so we used $\beta = 0.5/\rho(A)$ instead.

6.7 DeepWalk+HOPE

Looking at the cross-tables of clusters against Rao's labels, we noticed that DeepWalk separated A1 well but mixed up the B classes, while HOPE separated B1, B2 and B3 well but merged A1 and A2 together. Because of this we tried combining the two. Each embedding is first normalised to unit length and then the two are concatenated, giving 200 dimensions. This was decided before running and it was tried only once.

6.8 Results

All the results on the 26,226 labelled bins are given in Table 4, and Figure 2 shows the ARI against Rao for both graphs.

Table 4: Results on the 26,226 Rao A1–B3 bins. ARI/NMI against Rao (R) and against the published SCI labels (S). Lowest-inertia solution of three seeds, 100 restarts each.

Graph	Method	Sil.	DB	ARI R	NMI R	ARI S	NMI S
INTERKR	LINE, concatenated	0.005	8.34	0.049	0.060	0.041	0.048
	LINE, per-order L2, 1000M	0.076	3.33	0.443	0.487	0.268	0.311
	LINE, joint (25M)[†]	0.545	0.505	0.409	0.466	0.312	0.356
	Weighted DeepWalk	0.202	1.49	0.341	0.448	0.258	0.334
	HOPE	0.093	2.86	0.550	0.543	0.305	0.334
	DeepWalk + HOPE	0.093	2.70	0.526	0.544	0.306	0.357
Pair-O/E	LINE, concatenated	−0.000	8.77	0.195	0.290	0.148	0.204
	LINE, per-order L2, 1000M	0.058	3.46	0.551	0.560	0.352	0.388
	LINE, joint (25M)	–	–	0.329	0.437	0.264	0.327
	Weighted DeepWalk	0.274	1.30	0.428	0.527	0.324	0.380
	HOPE	0.091	2.69	0.616	0.624	0.350	0.407
	DeepWalk + HOPE	0.104	2.57	**0.633**	**0.646**	0.362	0.421
	Published SCI vs Rao			0.503	0.567		

[†]Two clusters hold only 47 bins, which inflates its silhouette, Davies–Bouldin and ARI.

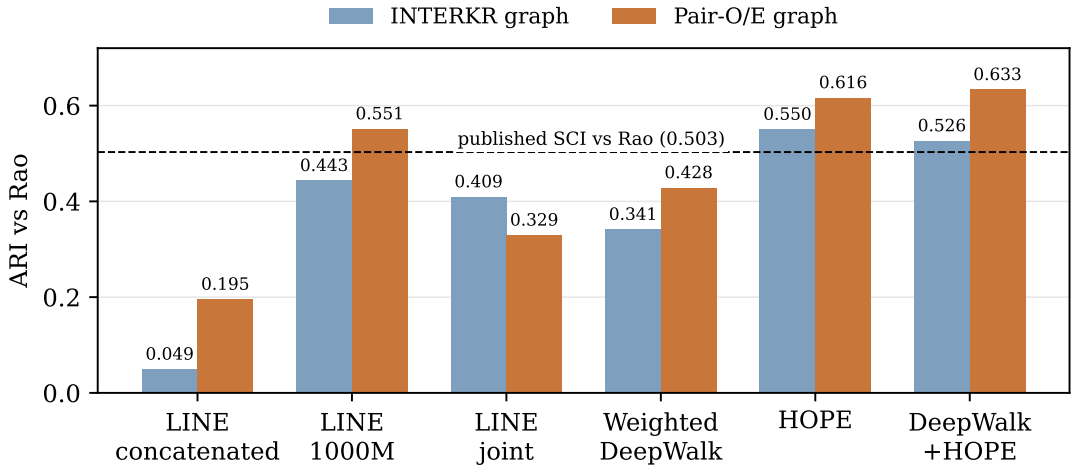


Figure 2: ARI against Rao's labels. The dashed line is the published SCI labels' agreement with Rao.

HOPE and DeepWalk+HOPE perform the best on both graphs. Their scores are very close to each other and it is not possible to say that one is better than the other. Weighted DeepWalk has a lower ARI, and combining DeepWalk with HOPE gave only a small improvement.

Internal scores do not match ARI. Across 11 clusterings, the correlation between silhouette and ARI was only about 0.08. For example, weighted DeepWalk has the highest silhouette among the well-formed clusterings, but its ARI is lower than that of HOPE. This is the same kind of score that made DeepWalk look like the best method on the demo data, so we did not use internal scores for choosing between methods.

Are the other methods actually better than LINE? To check this, we used paired differences with chromosome resampling (Table 5). Every method beats concatenated LINE by a large margin. When compared with the improved LINE (per-order L2, 1000M) though, the situation is different. HOPE and DeepWalk+HOPE still win, but only by around 0.06 to 0.11 ARI. On the pair-O/E graph, the interval for DeepWalk+HOPE starts at +0.004 and the interval for HOPE includes zero. Weighted DeepWalk alone does not beat the improved LINE. This shows that most of SCI's weakness comes from the way the two LINE orders are joined, and not from LINE as such.

Table 5: ARI gain against Rao over the LINE variants (first minus second), with 95% chromosome resampling intervals. * = interval excludes zero.

Comparison	INTERKR	Pair-O/E
HOPE − LINE concatenated	$+0.501^*$	$+0.421^*$
DeepWalk+HOPE − LINE concatenated	$+0.477^*$	$+0.438^*$
DeepWalk − LINE concatenated	$+0.291^*$	$+0.233^*$
HOPE − LINE per-order L2 (1000M)	$+0.107^*$ [0.071, 0.150]	$+0.064$ [−0.015, 0.151]
DeepWalk+HOPE − LINE per-order L2 (1000M)	$+0.083^*$ [0.057, 0.122]	$+0.081^*$ [0.004, 0.179]
DeepWalk − LINE per-order L2 (1000M)	−0.102 [−0.202, 0.033]	-0.123^* [−0.186, −0.014]
HOPE − LINE joint	+0.141 [−0.002, 0.251]	$+0.287^*$ [0.232, 0.345]
DeepWalk+HOPE − LINE joint	$+0.117^*$ [0.013, 0.206]	$+0.304^*$ [0.256, 0.355]

7 Pair-O/E Normalisation

7.1 Motivation

On the INTERKR graph, the best performing methods had a high NMI with chromosome identity: 0.199 for HOPE, 0.192 for LINE 1000M and 0.154 for DeepWalk+HOPE. This is 1.5 to 2 times the value of 0.100 for Rao's own labels, which means that some of their agreement with Rao may be coming from chromosome structure rather than from sub-compartments. One likely reason is that entire chromosome pairs differ in their average contact level, so the embedding partly ends up learning "which chromosome pair this is" instead of "which sub-compartment this is".

7.2 Method

For every chromosome-pair block in the contact matrix, each entry is divided by the mean of that block over valid bins. This gives an observed/expected value for trans contacts and removes the overall contact level of each chromosome pair. The same edge rule is then applied (value ≥ 1, weight $\ln(\text{value})$). This kind of per-pair trans expected is a standard correction and is also available in tools like cooltools. As far as we could find, it has not been applied before graph embedding for sub-compartments, but we consider it a standard correction and not a new method. The pair-O/E graph has 37% fewer edges than INTERKR (Table 2). This single variant was decided before running and tested only once.

7.3 Results

Pair-O/E improved the ARI of every method whose clustering is well-formed (Table 4, Figure 2): concatenated LINE from 0.049 to 0.195, LINE 1000M from 0.443 to 0.551, DeepWalk from 0.341 to 0.428, HOPE from 0.550 to 0.616 and DeepWalk+HOPE from 0.526 to 0.633. With chromosome resampling, the improvement is clear for four out of the five methods, while for HOPE the interval includes zero. Joint LINE is left out of this list, since its INTERKR clustering has two outlier clusters (Section 6.4).

Figure 3 shows that pair-O/E does not work by making the clusters follow chromosomes. On the pair-O/E graph, the chromosome NMI of every method lies between 0.023 and 0.060, which is below Rao's value of 0.100. It did not go down for every method, since it stayed about the same for DeepWalk and went up slightly for concatenated LINE, but all values are still below Rao's.

On the pair-O/E graph, DeepWalk+HOPE and HOPE agree with Rao better than the published SCI labels, and this difference holds up under chromosome resampling: +0.130 [0.083, 0.184] for DeepWalk+HOPE and +0.113 [0.061, 0.167] for HOPE. On the INTERKR graph these differences are not clear, as the intervals include zero.

One unusual case was LINE order 1 alone at 1000M samples on pair-O/E, which gave an ARI of 0.099 (NMI 0.347) because one single cluster contained 67% of all nodes. This does not affect the per-order L2 result.

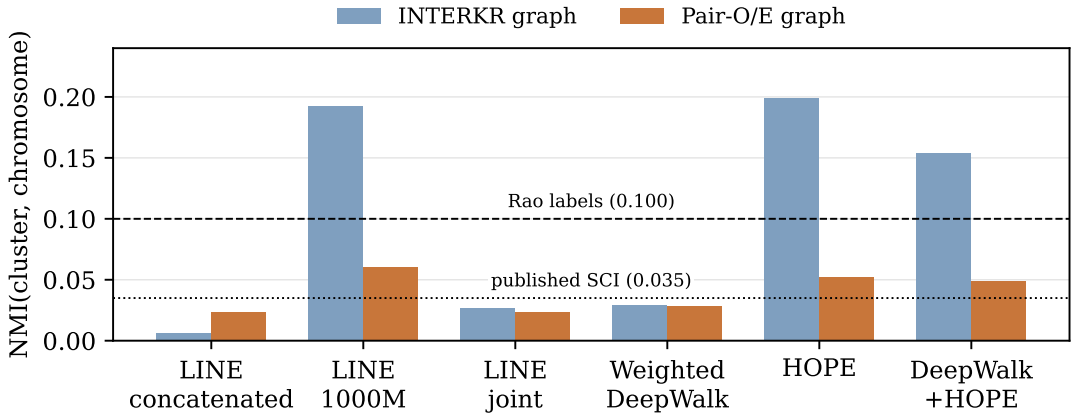


Figure 3: NMI between clusters and chromosome. Dashed: Rao's labels. Dotted: published SCI labels.

8 Biological Validation and Reproducibility

Agreeing more with Rao's labels does not necessarily mean being biologically better, especially since the SCI paper claims to improve upon Rao. So we also checked the clusterings against independent biological signals. The second half of this section describes our attempts to regenerate the published SCI labels.

8.1 Replication timing

We used replication timing data for GM12878 from ENCODE (UW Repli-seq, hg19). For every 100-kb bin, the mean signal was taken, and for every clustering we computed η^2, which is the fraction of the variance in the signal that is explained by the five clusters. Since this does not need the clusters to be matched to A1–B3, it can be used for any labelling. All files were verified using their md5 checksums and hg19 chromosome lengths, and the plan was fixed before any scores were computed.

8.2 Histone marks

The same approach was used for three histone marks from ENCODE: H3K27me3 (Polycomb, associated with B1), H3K9me3 (heterochromatin) and H3K4me3 (active promoters).

8.3 Results

The replication timing scores are given in Table 6 and Figure 4.

Replication timing turned out to be the useful signal here. On the pair-O/E graph, DeepWalk+HOPE, HOPE and LINE 1000M reach values of 0.64 to 0.65. Under chromosome resampling these cannot be distinguished from Rao's value of 0.647, and they are higher than that of the published SCI labels (0.582). Pair-O/E increased the timing score for every method, just like it increased the ARI. Chromosome identity alone explains only 0.081, so these scores are not simply coming from the chromosomes.

Table 6: Replication-timing η^2 (share of variance explained by the labels).

Labelling	INTERKR	Pair-O/E
LINE, concatenated	0.130	0.479
LINE, per-order L2, 1000M	0.540	0.642
LINE, joint (25M)	0.638	0.679
Weighted DeepWalk	0.703	0.712
HOPE	0.568	0.638
DeepWalk + HOPE	0.632	0.654
Rao labels	0.647	
Published SCI labels	0.582	
Chromosome (22 groups)	0.081	

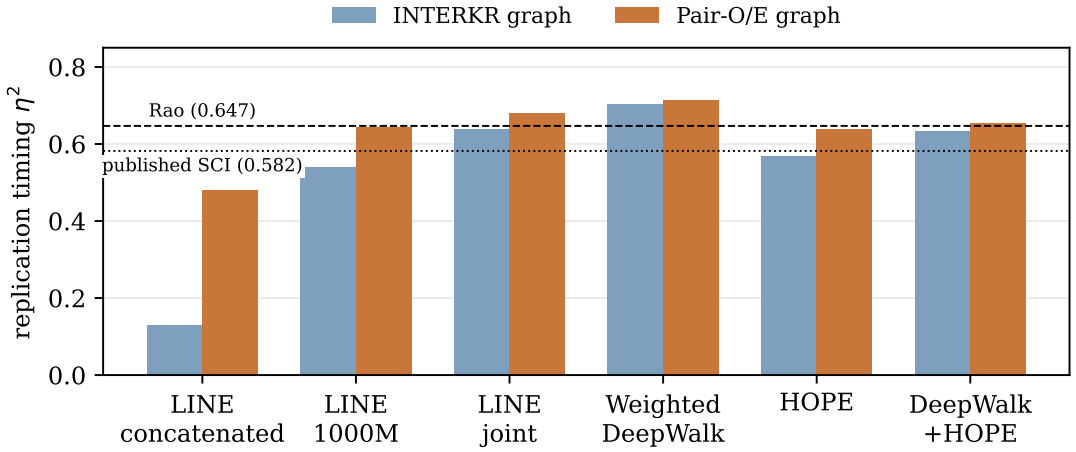


Figure 4: Replication-timing η^2. Dashed: Rao's labels. Dotted: published SCI labels.

Weighted DeepWalk gets the highest timing score (0.70 to 0.71), even though its ARI is lower than HOPE. Replication timing mostly follows one smooth gradient from early to late, so any clustering that follows this gradient will score well, even if its groups do not correspond to A1–B3. Therefore a high timing score on its own does not mean the sub-compartments are better.

The three histone marks explain almost nothing for any of the labellings, including Rao's ($\eta^2 \leq 0.019$ for H3K27me3, ≤ 0.005 for H3K9me3 and ≤ 0.041 for H3K4me3), so they are not useful for separating the methods at 100-kb resolution.

8.4 Published SCI labels

Our LINE runs did not match the published SCI labels (ARI of about 0.04 for the concatenated mode), and this was the part of the project that took the most effort. Before starting the tests, we fixed a rule: an ARI of 0.7 or above against the published labels would count as a match, 0.4 to 0.7 as a partial match, and anything below 0.4 as no match. All the attempts ended up as "no match":

- **LINE settings.** SCI's code calls LINE with only the order and number of samples, so the remaining settings are LINE's defaults, which are the same as ours.
- **The three LINE modes.** The best result was 0.352 (per-order L2, 1000M, pair-O/E).
- **Only the 26,226 labelled bins.** Clustering without the 517 extra bins did not help (best value 0.263).

- **Other normalisations and read filters.** We built graphs using genome-wide KR at MAPQ≥30, genome-wide KR at MAPQ≥0, and per-chromosome KR at MAPQ≥0 (which is the input closest to what the authors used). These graphs have between 225 and 278 million edges, and none of them gave an ARI above 0.25 against the published labels.

We also tested the paper's claim that SCI's labels are more compact than Rao's. Within our embeddings, Rao's labels were tighter in nearly all cases (lower Davies–Bouldin in all 9 embeddings and higher silhouette in 8 out of 9). This does not disprove the claim, as the embedding used in the paper is not available.

8.5 Authors' container

We ran the authors' Docker image on the cluster using Apptainer. By comparing the demo values with the archive, we found that the demo data is exactly the unfiltered (MAPQ≥0) contacts with per-chromosome KR normalisation, which is different from the genome-wide KR mentioned in the authors' conversion script. On three chromosome pairs, the demo keeps about 3% of the entries in the archive (3.03%) at every contact strength, which means it is a uniform random 3% sample. This is why the demo graph (8.35 million edges) is much smaller than the full graph built from the same type of input (276 million edges).

We also noticed that in SCI's code the command for the single-order mode is missing a space (`-order 1-samples`), so in that mode the settings may not be passed to LINE correctly.

8.6 Repeated runs

Table 7 shows the results of running SCI's own code on its own demo file, compared with the demo result that is saved inside the container.

Table 7: SCI's own code on its own demo file, compared with the demo result saved in the container.

Run (authors' code)	ARI vs saved result	Cluster sizes
`-o both -s 25` (first run)	0.003	10283, 8444, 8247, 2, 1
`-o both -s 25` (repeat)	0.006	7639, 6783, 6576, 5978, 1
`-o 1 -s 25`	0.008	balanced
`-o both -s 1`	0.006	balanced

From this we observed three things:

1. With both orders, the authors' code collapses in the same way as ours (clusters with only 1 and 2 bins). So the collapse comes from SCI's method itself and not from our implementation.
2. Two runs with exactly the same settings agree with each other at an ARI of only 0.041, which means the authors' pipeline does not give the same result when run again. We do not know the exact reason. One possible reason, which we have not tested yet, is that K-means in SCI's clustering script is not seeded.
3. None of the settings we tried was able to reproduce the result saved in the container.

8.7 Interpretation

Since the authors' own container cannot reproduce its own saved demo result and gives different results every time it is run, we do not expect to be able to regenerate the published labels with any input, and we

stopped the reproduction work at this point. This also explains our demo results from Section 5.1: the graph building step is deterministic and matched, but the clustering step in the original pipeline is not stable.

9 Discussion

9.1 SCI and LINE

The main issue with SCI is not LINE itself, but the way its two orders are combined. The order-2 vectors are much longer than the order-1 vectors, so once they are concatenated, the noisy order-2 part dominates the distances. Normalising each order first, as is done in the original LINE toolkit, fixes most of this. The same collapse can also be seen when running the authors' code on their own demo data.

9.2 Embedding methods

HOPE and DeepWalk+HOPE gave the best agreement with Rao on both graphs, but since their scores are very close, neither can be called the single best method. Weighted DeepWalk looked like the best method on the demo data based on internal scores, and on the full data it has the highest silhouette and replication timing score as well. However, its ARI is lower than HOPE's and it does not beat the improved LINE. This taught us that internal scores and a single biological signal can point towards a different winner than agreement with labels, which is why we consider several scores together.

9.3 Normalisation

In some cases, the pair-O/E correction helped more than switching to a different embedding method. For instance, LINE 1000M on pair-O/E (0.551) does as well as HOPE on INTERKR (0.550). This suggests that the way the graph is constructed is as important as the way it is embedded. Our explanation is that pair-O/E removes the differences in overall contact level between chromosome pairs, although we have not tested this directly.

9.4 Biological interpretation

At 100-kb resolution, replication timing was the only independent signal that turned out to be informative. Our best clusterings reach the same level as Rao's labels on it, but do not go above it. So we cannot claim that our clusters are biologically better than Rao's, only that they are equally good on this signal and better than the published SCI labels.

9.5 Implications

To be honest, the improvement is not very large. Most of the big jump over SCI comes from fixing the way the orders are merged, and once LINE is fixed the other methods only add about 0.06 to 0.11 ARI. The more useful findings might actually be the negative ones, that is, the concatenated mode collapses, the published labels cannot be regenerated, and the authors' pipeline is not deterministic. These are important for anyone who plans to use SCI's labels or code.

10 Limitations

- Rao's labels are only a reference and not the ground truth, so higher agreement with them does not prove that a clustering is better.
- All the work is done on a single cell line (GM12878) and at a single resolution (100 kb).
- Some choices were made by us, such as the INTERKR normalisation (since the paper does not mention one), the value of β and the dimension in HOPE, and the DeepWalk settings. None of these were tuned.
- The ideas of DeepWalk+HOPE and pair-O/E came from looking at our own results (the cross-tables and chromosome NMI). Both were written down before running and tested only once, but they are still influenced by the data.
- The number of LINE samples and the way of merging were changed after seeing the LINE scores. For this reason we report LINE with the paper's settings as the main LINE result and treat the others as sensitivity checks.
- Bins with low coverage can get inflated INTERKR values.
- We did not measure how much the embeddings change from one run to another, and LINE does not have an option to set the seed.
- Our reproduction of SCI on the full data does not match the published labels.

11 Future Work

A few checks are still pending for the first part of the project. These include SCI's own main test (ChIA-PET loop ratios within and between sub-compartments), checking whether K-means in SCI is seeded, and selecting settings on odd chromosomes while reporting results on even chromosomes. The main focus from here on, however, will be the second part of the project.

11.1 Genome data structure

This part comes directly from the problems we faced in the first part. For a single cell line, the inter-chromosomal archive is 36 GB when compressed and 224 GB when uncompressed, and a single graph built from it takes 8.6 to 11 GB. In almost every experiment, a large amount of data had to be read or extracted just to use a small part of it, such as one resolution, one read filter, one normalisation or a few chromosome pairs. The goal is to design a data structure that stores Hi-C contact data in a compact way and allows fast access to any part of it, without having to unpack everything.

11.2 Query processor

On top of this data structure we plan to build a query processor that can answer queries like "give the contacts between these two regions", "give all contacts of this bin" or "give this normalised chromosome-pair block". It should only read the part of the data that is needed for a particular query. The input formats, the types of queries, the programming language and the performance targets will be decided together with our supervisor.

12 Conclusion

In this project we reproduced SCI's pipeline on the demo dataset, where the graph matched the authors' code and sci-DNN matched the paper, and then rebuilt SCI on the full GM12878 data. The main conclusions are:

- SCI's way of combining the two LINE orders leads to a collapse, because the order-2 vectors are about 70 times longer. The same collapse is seen in the authors' own code, and normalising each order before joining fixes most of it.
- HOPE and DeepWalk+HOPE gave the best results, and the pair-O/E correction improved every well-formed clustering without making the clusters follow chromosomes. On the pair-O/E graph, DeepWalk+HOPE reaches an ARI of 0.633 against Rao, compared to 0.503 for the published SCI labels.
- The improvement is modest. Compared to an improved LINE, the gain is only about 0.06 to 0.11 ARI, and on replication timing our best methods match Rao's labels but do not beat them.
- We were not able to reproduce the published labels. The authors' container does not give back its own saved demo result and gives different results when run again.

In the next part of the project, we will design a data structure and query processor for Hi-C contact data, based on the data handling problems faced during this work.

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